

Pratyush Singh

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SUMMARY

Computer Science Engineering student specializing in Cloud Computing and Automation, with hands-on experience in full-stack development, cloud infrastructure, and AI-driven applications. Proficient in developing web applications using React.js, Node.js, Python, and REST APIs, with experience in databases, cloud platforms, and containerization. Strong foundation in software development, problem-solving, data structures, and cloud technologies, with experience developing and deploying academic and personal projects.

EDUCATION

VIT Bhopal University

Bhopal, Madhya Pradesh

B.Tech in Computer Science and Engineering (Cloud Computing and Automation)

CGPA: 8.00/10

Bhavan's Kesari Devi Kanoria Vidya Mandir

Sonebhadra, Uttar Pradesh

Class-X: 92.5%

Class-XII: 73.5%

TECHNICAL SKILLS

Languages : Python, C++, JavaScript, SQL, HTML, CSS

Frameworks : React.js, Node.js, Express.js, Flask, FastAPI

Databases : MongoDB, MySQL, PostgreSQL

Cloud & DevOps : AWS, Google Cloud Platform, Docker, Git, GitHub

PROJECTS

DevSync | *React, Node.js, Express, MongoDB, Socket.IO, JWT*

May 2026 – Present

- Architected a multi-tenant SaaS platform with a 4-level Organization → Team → Project → Task hierarchy spanning 40+ REST endpoints across 10+ modular services, deriving authorization from each resource to eliminate IDOR vulnerabilities.
- Engineered stateless JWT authentication using 15-minute access tokens and 7-day httpOnly refresh-token rotation, enabling silent session restore via Axios interceptors (401 → refresh → retry) and bcrypt-hashed credentials.
- Hardened role-based access control (owner/member) through layered Express middleware, validated by 50+ automated authorization tests with zero access-control failures.
- Orchestrated a real-time layer with Socket.IO powering 3 live features — notifications, drag-and-drop Kanban sync, and team chat — over WebSockets scoped to isolated per-organization rooms.
- Deployed the full-stack application to the cloud (Render + Vercel + MongoDB Atlas), securing all API routes with production CORS, rate limiting, and Zod schema validation.

Employee Management System | *React.js, Node.js, Express.js, MongoDB, REST APIs*

Aug 2025

- Built a full-stack employee management system covering employee records, departments, projects, tasks, leave, and payroll operations.
- Implemented authentication and role-based access control for 3 user roles: Admin, HR, and Employee, with role-specific dashboards and permissions.
- Developed RESTful APIs using Node.js and Express.js for employee management, project tracking, task assignment, and organizational workflows.
- Designed MongoDB schemas and database operations for storing and managing users, employees, projects, tasks, and company-related data.

CERTIFICATIONS

NPTEL: Introduction to Internet of Things Certificate

Microsoft: Azure Data Fundamentals Certificate

AWS: Certified Cloud Practitioner (CLF-C02)

AWS: Certified Cloud Solutions Architect–Associate (SAA-C03)

ACHIEVEMENTS

- Achieved a 5-Star rating in C++ on HackerRank, demonstrating strong proficiency in C++ programming, Data Structures, Algorithms, and Problem Solving.
- Solved 300+ coding problems across LeetCode and other programming platforms and achieved a LeetCode ranking under 500,000, demonstrating strong problem-solving skills in Data Structures and Algorithms.